Worksheet 3

Dispersion and concentration

Complete exercises

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1. Core applications

Master the essential definitions and calculations from the course.

Core application 25 min

Exercise 4 — Medial and cumulative relative masses

The spending of 40 customers (€) is grouped below. Approximate class monetary mass by $n_i \times$ class midpoint.

Classe / Class	[10,20[	[20,30[	[30,40[	[40,50[	[50,60[
n_i	8	12	10	6	4

1. Compute midpoints, masses, cumulative masses and relative cumulative mass shares.
2. Interpolate the medial in the class containing half of total mass.
3. Compute the frequency median and compare the two statistics.

Core application 20 min

Extra 1 — Dispersion from a frequency table

The number of weekly complaints per store is distributed as follows:

Complaints x	0	1	2	3	4
Stores n	4	8	10	6	2

1. Calculate the mean, population variance and standard deviation.
2. Calculate the coefficient of variation.
3. Interpret every result in context.

Build columns nx and nx^2. Use $E(X^2) - [E(X)]^2$ for the population variance.

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 35 min

Exercise 2 – Dispersion of grouped data

Using the 210-customer distribution from Worksheet 2, calculate $\sum n_i c_i$, $\sum n_i c_i^2$, mean, population variance, standard deviation, coefficient of variation and IQR. Compare the information provided by standard deviation and IQR.

Assessment / exam practice 30 min

Exercise 3 – Lorenz curve and Gini coefficient

Income values are $10, 12, 18, 25, 35$. Calculate cumulative population and income shares, construct the Lorenz curve, calculate the area by trapezoids, derive the Gini coefficient and explain why Gini measures concentration without explaining its causes.

3. Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice 30 min

Exercise 1 – Boxplot, potential outliers and variance choice

Consider

$$5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 11, 20.$$

Use the median-of-halves convention for quartiles.

1. Calculate the five-number summary.
2. Calculate the range and IQR.
3. Calculate the fences $Q_1 - 1.5IQR$ and $Q_3 + 1.5IQR$ and identify potential outliers.
4. Calculate the mean and mean absolute deviation.
5. Calculate population variance and standard deviation if the 12 observations are the entire population.
6. Calculate sample variance and standard deviation if they are a sample.

Advanced practice 25 min

Extra 2 — Comparing concentration and robustness

Two markets have the following ordered firm market shares:

Market A	10%	15%	20%	25%	30%
Market B	2%	8%	10%	20%	60%

1. Construct cumulative shares for each market.
2. Explain which Lorenz curve is farther from equality.
3. For each market calculate the range and discuss why it is insufficient.
4. State a defensible concentration conclusion.

With five equal-weight firms, cumulative population shares increase by 20 percentage points.